PBUF Science Run Comparison Report

Generated 2025-11-02 08:39:22 UTC — Models: LCDM, PBUF

1. Aggregated Overview

1.1 Dataset Diagnostics

Dataset	χ^2 LCDM	χ^2 PBUF	$\Delta\chi^2$ (pbuf-lcdm)	AIC LCDM	AIC PBUF	ΔAIC	BIC LCDM	BIC PBUF	ΔBIC
BAO_ANISO	63.0181	62.9832	-0.0349	0	0	0	0	0	0
CC	19.5849	19.7652	0.1803	0	0	0	0	0	0
CMB	0.9523	0.9506	-0.0017	0	0	0	0	0	0
RSD	11.5679	11.4638	-0.1042	0	0	0	0	0	0

1.2 Global Model Totals

Model	χ^2 total	AIC total	BIC total	Reduced χ^2	# Data	# Params	# Runs
LCDM	883.342330	891.342330	913.273058	0.498219	1777	4	1
PBUF	882.478509	906.478509	972.270691	0.499988	1777	12	1

1.3 Model Preference

Metric	Value	Preferred Model
ΔAIC (PBUF-LCDM)	15.1362	LCDM
ΔBIC (PBUF-LCDM)	58.9976	LCDM

Positive ΔAIC/ΔBIC values favour LCDM; negative values favour PBUF. Magnitudes > 10 typically signal decisive evidence.

1.4 Average Best-Fit Parameters

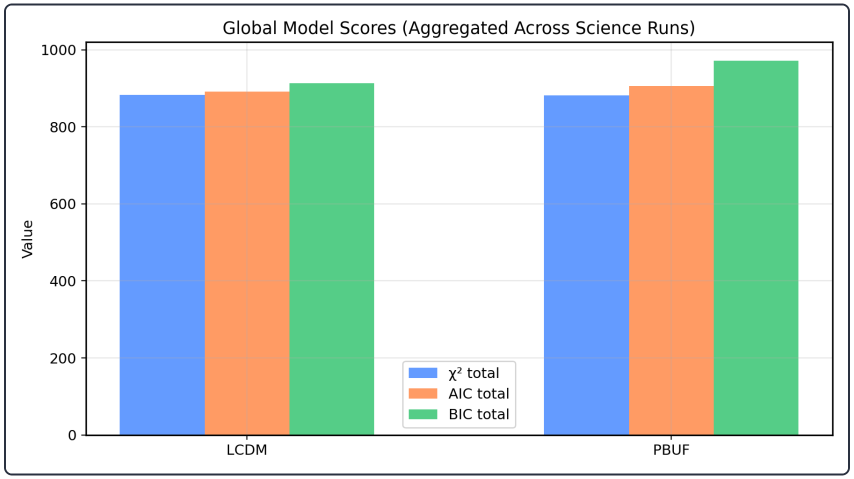
LCDM Parameters (run-average)

H0	73.996722
Ok0	0
Om0	0.239080
Or0	9.20e-05

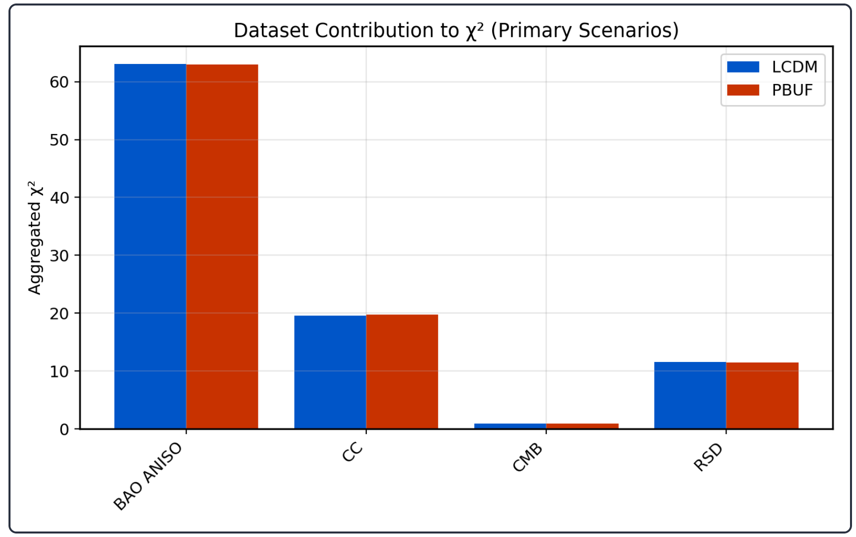
PBUF Parameters (run-average)

H0	73.995228
Ok0	0
Ol0	0
Om0	0.240034
Or0	9.20e-05
Rmax	5.00e+06
alpha	0.022776
eps0	0.909768
k_sat	0.978691
n_R	0
n_alpha	0.800000
n_eps	-0.500000

1.5 Aggregated Plots



Aggregated model totals (χ^2, AIC, BIC)



Aggregated dataset χ^2 contributions

2. Science Run — 20251101_232513_v10_full_run

Metadata

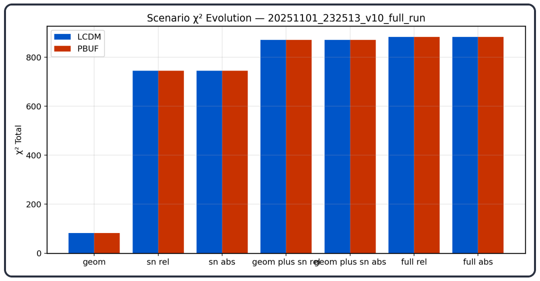
- **Directory**: /home/fabian/PBUF4/data/science_runs/20251101_232513_v10_full_run

Scenario Summary

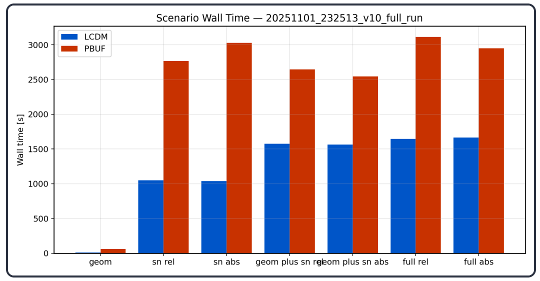
Scenario	Model	χ^2	AIC	BIC	Runtime [s]	Phase-6a	ΔAIC (PBUF-LCDM)	ΔBIC (PBUF-LCDM)
geom	LCDM	81.8590	89.8590	98.1008	11.8189	✅ pass	16.0432	32.5267
	PBUF	81.9022	105.9022	130.6275	61.1503	✅ pass (enforced)		
sn_rel	LCDM	745.4017	753.4017	775.1576	1050.2222	✅ pass	16.0000	59.5118

Scenario	Model	χ^2	AIC	BIC	Runtime [s]	Phase-6a	ΔAIC (PBUF-LCDM)	ΔBIC (PBUF-LCDM)
	PBUF	745.4017	769.4017	834.6694	2767.9485	✅ pass (enforced)		
sn_abs	LCDM	745.4017	753.4017	775.1576	1036.3124	✅ pass	16.0000	59.5118
	PBUF	745.4017	769.4017	834.6694	3028.9221	✅ pass (enforced)		
geom_plus_sn_rel	LCDM	870.9649	878.9649	900.8549	1574.2658	✅ pass	16	59.7800
	PBUF	870.9649	894.9649	960.6349	2646.0992	✅ pass (enforced)		
geom_plus_sn_abs	LCDM	870.9649	878.9649	900.8549	1565.1141	✅ pass	16	59.7800
	PBUF	870.9649	894.9649	960.6349	2543.9280	✅ pass (enforced)		
full_rel	LCDM	883.3423	891.3423	913.2731	1646.1679	✅ pass	15.1362	58.9976
	PBUF	882.4785	906.4785	972.2707	3114.7905	✅ pass (enforced)		
full_abs	LCDM	883.3423	891.3423	913.2731	1665.0143	✅ pass	15.0894	58.9509
	PBUF	882.4317	906.4317	972.2239	2950.4286	✅ pass (enforced)		

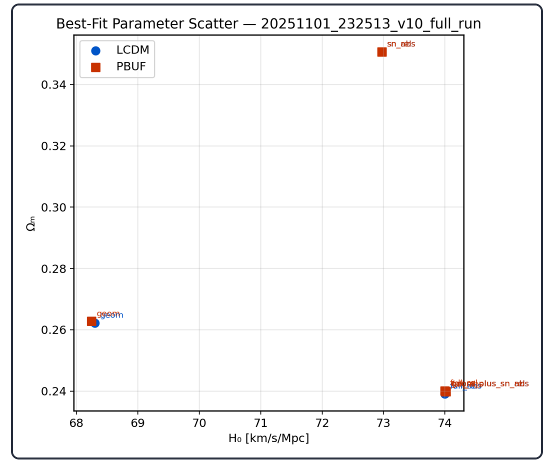
Plots



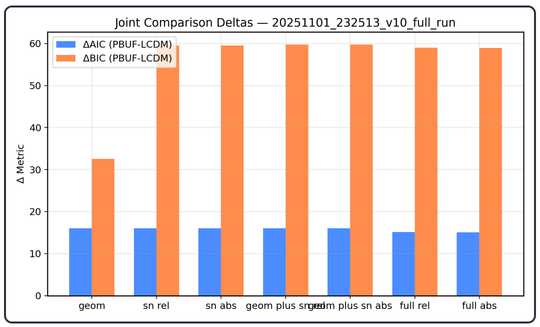
Scenario χ^2 totals



Scenario wall time



Best-fit H_0 vs Ω_m



Joint ΔAIC / ΔBIC

3. Appendices

Appendix A — Raw Statistics Object

▶ Raw statistics object (click to expand)

Appendix B — Notes

Methodology Notes

- Aggregates combine the primary scenario of each science run (highest data volume).
- ΔAIC/ΔBIC are computed as ($MODEL_2 - MODEL_1$); positive values favour $MODEL_1$.
- Phase-6a status reflects the guard-rail enforcement during the coordinate optimisation.
- Per-run parameter cards display the mean of best-fit values across runs.
- Plots are generated via `reports.plotter`; regenerate them after new science runs.